

Assignment No.12

Problem Statement : Design and Develop MongoDB Queries using aggregation and indexing with suitable example using MongoDB.

Program :

```
show dbs
use bill
db.pay.insert({Cust_ID:"A123",Product:"Milk",Amount:40,Status:"P"});
db.pay.insert({Cust_ID:"A123",Product:"Parle_G",Amount:50,Status:"NP"});
db.pay.insert({Cust_ID:"A123",Product:"Lays Chips",Amount:40,Status:"P"});
db.pay.insert({Cust_ID:"B123",Product:"Mentos",Amount:10,Status:"P"});
db.pay.insert({Cust_ID:"B123",Product:"Maggie",Amount:60,Status:"NP"});
db.pay.find()
db.pay.mapReduce(function() {emit(this.Cust_ID,this.Amount);},function(key,values){return
Array.sum(values)}},{query: {"Status":"NP"},out:"Bill_Amount"})
var mapFunc1=function() {emit(this.Cust_ID,this.Amount);};
var reduceFunc1=function(keyCustID,valuePrices) {return Array.sum(valuePrices);};
db.pay.mapReduce(mapFunc1,reduceFunc1,{out:"Map"})
db.Bill_Amount.find()
db.Map.find()
```

Output :

```
test> switched to db bill
bill> DeprecationWarning: Collection.insert() is deprecated. Use insertOne, insertMany, or
bulkWrite.
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('6904f488201dada35ece5f4f') }
}
bill> {
  acknowledged: true,
  insertedIds: { '0': ObjectId('6904f488201dada35ece5f50') }
```

```
}
```

```
bill> {
```

```
  acknowledged: true,
```

```
  insertedIds: { '0': ObjectId('6904f488201dada35ece5f51') }
```

```
}
```

```
bill> {
```

```
  acknowledged: true,
```

```
  insertedIds: { '0': ObjectId('6904f488201dada35ece5f52') }
```

```
}
```

```
bill> {
```

```
  acknowledged: true,
```

```
  insertedIds: { '0': ObjectId('6904f488201dada35ece5f53') }
```

```
}
```

```
bill> [
```

```
{
```

```
  _id: ObjectId('6904f488201dada35ece5f4f'),
```

```
  Cust_ID: 'A123',
```

```
  Product: 'Milk',
```

```
  Amount: 40,
```

```
  Status: 'P'
```

```
},
```

```
{
```

```
  _id: ObjectId('6904f488201dada35ece5f50'),
```

```
  Cust_ID: 'A123',
```

```
  Product: 'Parle_G',
```

```
  Amount: 50,
```

```
  Status: 'NP'
```

```
},
```

```
{
```

```
  _id: ObjectId('6904f488201dada35ece5f51'),
```

```

    Cust_ID: 'A123',
    Product: 'Lays Chips',
    Amount: 40,
    Status: 'P'
  },
  {
    _id: ObjectId('6904f488201dada35ece5f52'),
    Cust_ID: 'B123',
    Product: 'Mentos',
    Amount: 10,
    Status: 'P'
  },
  {
    _id: ObjectId('6904f488201dada35ece5f53'),
    Cust_ID: 'B123',
    Product: 'Maggie',
    Amount: 60,
    Status: 'NP'
  }
]

```

bill> DeprecationWarning: Collection.mapReduce() is deprecated. Use an aggregation instead.

See <https://mongodb.com/docs/manual/core/map-reduce> for details.

```
{ result: 'Bill_Amount', ok: 1 }
```

bill>

bill>

```
bill> { result: 'Map', ok: 1 }
```

```
bill> [ { _id: 'A123', value: 50 }, { _id: 'B123', value: 60 } ]
```

```
bill> [ { _id: 'A123', value: 130 }, { _id: 'B123', value: 70 } ]
```